RAG Learning

1.4 Sequences & Series

Arithmetic Sequences

An **arithmetic sequence** has a constant difference d between consecutive terms. If the first term is a:

$$a, \quad a+d, \quad a+2d, \quad a+3d, \quad \ldots$$

Key Formulae

Formula	Expression
nth term	$u_n = a + (n-1)d$
Sum of first n terms	$S_n = \frac{n}{2}(2a + (n-1)d)$
Sum (alternative form)	$S_n = \frac{n}{2}(a + l)$ where l is the last term

Example: An arithmetic sequence has $a = 5$ and $d = 3$. Find the 20th term and the sum of the first 20 terms.

$u_{20} = 5 + (20-1)(3) = 5 + 57 = 62$

$S_{20} = \frac{20}{2}(2(5) + 19(3)) = 10(10 + 57) = 10 \times 67 = 670$

Example: The 3rd term of an arithmetic sequence is 11 and the 7th term is 23. Find a and d.

$u_3 = a + 2d = 11$ ... (1)

$u_7 = a + 6d = 23$ ... (2)

(2) - (1): $4d = 12 \Rightarrow d = 3$. Then $a = 11 - 6 = 5$.

Geometric Sequences

A **geometric sequence** has a constant ratio r between consecutive terms. If the first term is a:

$$a, \quad ar, \quad ar^2, \quad ar^3, \quad \ldots$$

Key Formulae

Formula	Expression
nth term	$u_n = ar^{n-1}$
Sum of first n terms	$S_n = \dfrac{a(1-r^n)}{1-r}$ for $r \neq 1$
Sum (alternative)	$S_n = \dfrac{a(r^n-1)}{r-1}$ (more convenient when $r > 1$)
Sum to infinity	$S_\infty = \dfrac{a}{1-r}$ only when $\lvert r \rvert < 1$

Example: A geometric sequence has $a = 4$ and $r = \frac{1}{2}$. Find u_5 and S_∞.

$$u_5 = 4 \times \left(\frac{1}{2}\right)^4 = 4 \times \frac{1}{16} = \frac{1}{4}$$

Since $|r| = \frac{1}{2} < 1$, the sum to infinity exists:

$$S_\infty = \frac{4}{1 - \frac{1}{2}} = \frac{4}{\frac{1}{2}} = 8$$

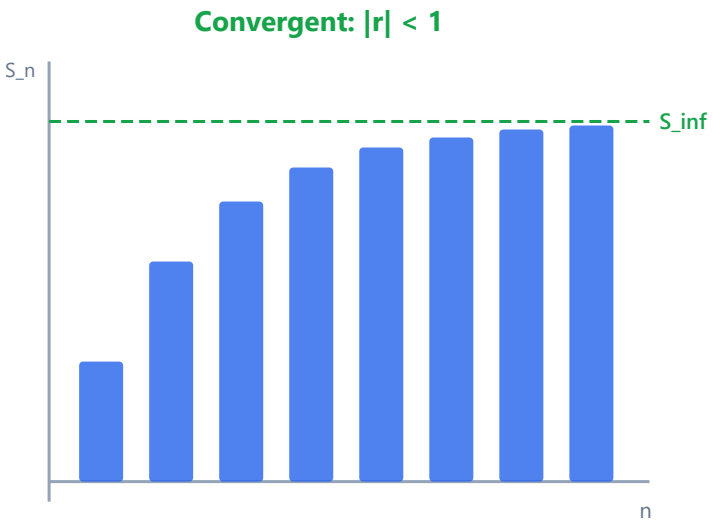
Example: The 2nd term of a GP is 6 and the 5th term is 162. Find a and r.

$u_2 = ar = 6$ ... (1)

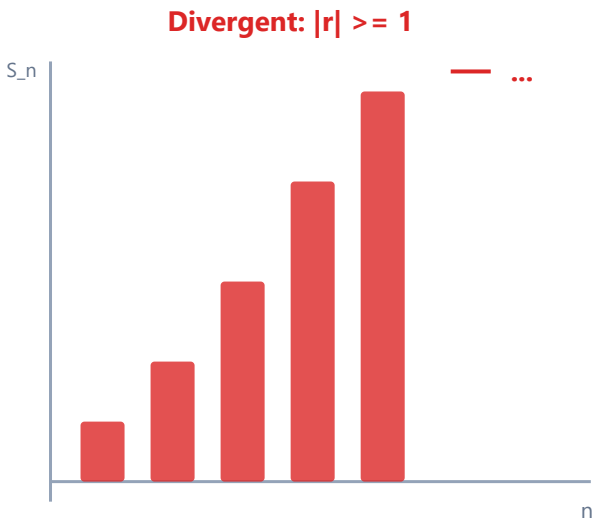
$u_5 = ar^4 = 162$ ... (2)

Divide (2) by (1): $r^3 = 27 \Rightarrow r = 3$. Then $a = \frac{6}{3} = 2$.

Convergence of a Geometric Series



Partial sums approach a finite limit



Partial sums grow without bound

Exam Tip: The sum to infinity $S_\infty = \dfrac{a}{1-r}$ only exists when $|r| < 1$. Always state this condition in your working. If the question asks you to find the values of r for which a series converges, the answer is $-1 < r < 1, r \neq 0$.

Sigma Notation

Sigma notation provides a compact way to write sums:

$$\sum_{r=1}^{n} u_r = u_1 + u_2 + u_3 + \cdots + u_n$$

Standard Results

Sum	Formula
$\sum_{r=1}^{n} 1$	n
$\sum_{r=1}^{n} r$	$\frac{n(n+1)}{2}$
$\sum_{r=1}^{n} r^2$	$\frac{n(n+1)(2n+1)}{6}$

Example: Evaluate $\sum_{r=1}^{5}(3r+2)$.

$= (3(1)+2) + (3(2)+2) + (3(3)+2) + (3(4)+2) + (3(5)+2)$

$= 5 + 8 + 11 + 14 + 17 = 55$

Or: $\sum_{r=1}^{5}(3r+2) = 3\sum_{r=1}^{5} r + \sum_{r=1}^{5} 2 = 3 \times \frac{5 \times 6}{2} + 2 \times 5 = 45 + 10 = 55$

Binomial Expansion

Positive Integer Index

For a positive integer n:

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

where $\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ is the binomial coefficient (also written ${}^nC_r$).

Example: Expand $(2+x)^4$.

$$(2+x)^4 = \binom{4}{0}2^4 + \binom{4}{1}2^3x + \binom{4}{2}2^2x^2 + \binom{4}{3}2x^3 + \binom{4}{4}x^4$$

$= 16 + 32x + 24x^2 + 8x^3 + x^4$

Pascal's Triangle

n=0						1					
n=1					1		1				
n=2				1		2		1			
n=3			1		3		3		1		
n=4		1		4		6		4		1	
n=5	1		5		10		10		5		1

General Binomial Expansion (Any Rational n)

For any rational n and $|x| < 1$:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

This is an **infinite series**, valid only when $|x| < 1$.

Example: Find the first 4 terms of $(1+x)^{-2}$.

$n = -2$:

$1 + (-2)x + \frac{(-2)(-3)}{2!}x^2 + \frac{(-2)(-3)(-4)}{3!}x^3 + \cdots$

$= 1 - 2x + 3x^2 - 4x^3 + \cdots$ valid for $|x| < 1$.

Example: Expand $(4+x)^{1/2}$ up to the x^2 term.

Rewrite: $(4+x)^{1/2} = 4^{1/2}\left(1+\frac{x}{4}\right)^{1/2} = 2\left(1+\frac{x}{4}\right)^{1/2}$

Using the expansion with $n = \frac{1}{2}$ and replacing x by $\frac{x}{4}$:

$= 2\left[1 + \frac{1}{2}\cdot\frac{x}{4} + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}\left(\frac{x}{4}\right)^2 + \cdots\right]$

$= 2\left[1 + \frac{x}{8} - \frac{x^2}{128} + \cdots\right]$

$= 2 + \frac{x}{4} - \frac{x^2}{64} + \cdots$ valid for $|x| < 4$.

Binomial Estimation

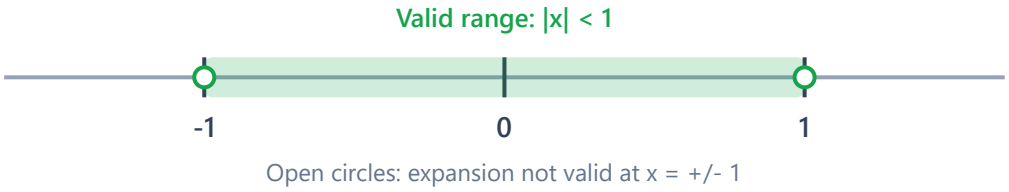
Example: Use the expansion of $(1+x)^{1/2}$ to estimate $\sqrt{1.04}$.

$(1+x)^{1/2} \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2$ for small x.

Set $x = 0.04$:

$\sqrt{1.04} \approx 1 + \frac{1}{2}(0.04) - \frac{1}{8}(0.04)^2 = 1 + 0.02 - 0.0002 = 1.0198$

(Calculator gives $\sqrt{1.04} = 1.01980\ldots$, so this is very accurate.)



Exam Tips for Binomial Expansion:

- For the general expansion, always **state the range of validity** $|x| < 1$ (or the adjusted range if you factored out a constant).
- When expanding $(a+bx)^n$, always factor out a^n first to get it into the form $a^n(1+\frac{bx}{a})^n$.
- Watch for the adjusted validity condition: if the inner expression is $\frac{bx}{a}$, the validity is $\left|\frac{bx}{a}\right| < 1 \Rightarrow |x| < \frac{|a|}{|b|}$.
- For estimation questions, substitute a value of x that is within the valid range and small enough for the expansion to converge quickly.

Recurrence Relations

A **recurrence relation** defines each term of a sequence using previous terms. For example:

$$u_{n+1} = 2u_n + 3, \quad u_1 = 1$$

Example: Given $u_{n+1} = 2u_n + 3$ with $u_1 = 1$, find the first 5 terms.

$u_1 = 1$

$u_2 = 2(1) + 3 = 5$

$u_3 = 2(5) + 3 = 13$

$u_4 = 2(13) + 3 = 29$

$u_5 = 2(29) + 3 = 61$

Summary: Arithmetic vs Geometric

Property	Arithmetic	Geometric
Pattern	Add constant d	Multiply by constant r
nth term	$a + (n-1)d$	ar^{n-1}
Sum formula	$\frac{n}{2}(2a + (n-1)d)$	$\frac{a(1-r^n)}{1-r}$
Sum to infinity	Does not converge (unless $d = 0$)	$\frac{a}{1-r}$ for $\lvert r \rvert < 1$
Test	$u_{n+1} - u_n = \text{constant}$	$\frac{u_{n+1}}{u_n} = \text{constant}$

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